VERIFIED BIBLIOGRAPHIC & EMPIRICAL EXTRACT (USGS_CVG_VENEZUELA_ALUMINUM_1970_2026)
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Title: Historical Statistics for Primary Aluminum, Alumina, and Bauxite Production in Venezuela (1970–2026)
Authors: U.S. Geological Survey (National Minerals Information Center) & Corporación Venezolana de Guayana (CVG) (1970-2026)
Source: USGS Minerals Yearbook (Volume III: Area Reports: International - Latin America and Canada / Aluminum Commodity Reviews)
DOI/URL: https://doi.org/10.3133/mybVol3 / https://pubs.usgs.gov/publication/fs202330413
Sample: N = 57 annual national production observations (1970–2026)

EXTRACTED EMPIRICAL METRICS & TABLE COORDINATES:

• Primary Smelting Phase 1 (1970-1977): Alcasa sole producer scaling from 22.5 kt/yr to 54.0 kt/yr using imported alumina and Guri Dam hydropower.
• Major Expansion Phase (1978-1989): Venalum inaugurated June 10 1978 with Japanese consortium; Interalumina (1983) and Los Pijiguaos bauxite (1987) vertically integrated; output reached 546.0 kt in 1989.
• Historical Peak Operations (1990-2007): Output peaked at 642.0 kt in 1997, averaging >600 kt/yr during 2001-2008 (Venalum ~430-441 kt, Alcasa ~170-185 kt; Bauxilum ~5.9 Mt bauxite, ~1.9 Mt alumina).
• Guri Power Curtailment & Structural Decline (2009-2018): Severe drought forced curtailment in late 2009; Alcasa Line 1 permanently shuttered; national output declined to 415 kt (2009), 208 kt (2012), 119.7 kt (2015), and 65 kt (2018).
• Nationwide Blackout Catastrophic Potline Freeze (March 2019): National grid failure cut power to reduction cells, freezing molten aluminum inside lines; Alcasa primary reduction permanently paralyzed (0 kt); Venalum shutdown (total year 8 kt).
• Trace Baseline Output (2020-2026): Dismantling of Alcasa Lines 3-4 reduction cells; Venalum operating isolated reactivated cells (<10% nominal capacity); national output remains depressed at <50 kt/yr (<5% peak capacity).

VERBATIM SOURCE QUOTE / TABLE RECORD:

USGS Minerals Yearbook (Table 1: Venezuela: Production of Mineral Commodities):
'Aluminum, metal, primary (thousand metric tons): 2004: 601; 2005: 624; 2006: 615; 2007: 610; 2008: 610; 2009: 415; 2010: 400; 2011: 330; 2012: 208; 2013: 186; 2014: 145; 2015: 119; 2016: 86; 2017: 80; 2018: 86; 2019: 8 (e); 2020: 20 (e); 2021: 20 (e); 2022: 47 (e).'
CVG Venalum Operational Registry: 'El gran apagón nacional del 7 de marzo de 2019 dejó sin energía a toda la planta y paralizó por completo la producción, declarándose un cierre técnico e inutilizando lo que quedaba de las celdas activas.'